APPENDIX I

GHANA—JOINT IMF AND WORLD BANK DEBT SUSTAINABILITY ANALYSIS[1]

Ghana's external debt distress risk rating is moderate, albeit close to the low risk category. The baseline scenario is contingent on continued macroeconomic stability and structural reforms that would lead to expansion of the private sector with resulting growth in GDP and exports. The analysis illustrates the importance of prudent debt management and judicious use of borrowed funds. The risk of debt distress related to total public debt—both external and domestic—is also moderate.

A. EXTERNAL DEBT SUSTAINABILITY ANALYSIS

Summary

The **baseline scenario** assumes (i) continued macroeconomic stabilization (prudent fiscal stance, single-digit inflation), (ii) perseverance with the pace of structural reforms (public financial management (PFM), reforms of state-owned enterprises (SOEs), including divestiture), and (iii) prudent borrowing strategy and efficient use of borrowed funds. It incorporates nonconcessional borrowing of US$700 million over the next three years and continued annual market access of about 1 percent of GDP for the rest of the projection period. In making these assumptions, the analysis attempts to be realistic with regard to commercial borrowing volumes and illustrates the value of prudence and the importance of the judicious use of nonconcessional funds.

In addition to the baseline scenario, three alternative scenarios are investigated:

- **Alternative scenario A** has the same borrowing path as in the baseline but lower growth of exports and GDP. This scenario examines whether the risk of debt distress would significantly increase should GDP growth and exports fail to accelerate.

- **Alternative scenario B** has higher nonconcessional borrowing than in the baseline but the same macroeconomic framework. The purpose is to examine the impact of higher borrowing on the risk of debt distress.

- **Alternative scenario C** has the higher nonconcessional borrowing path of Scenario B and the lower GDP and export growth of Scenario A. This scenario highlights the risks of high borrowing if growth of GDP and exports does not accelerate.

[1] Prepared by the IMF and World Bank staff in collaboration with the Ghanaian authorities.

1. Baseline Scenario

Macroeconomic Assumptions

The baseline scenario is contingent upon continued macroeconomic stability and structural reforms that will cause growth of GDP and exports to increase.

Box 1. Macroeconomic Assumptions for the Baseline Scenario

Real GDP grows at about 7 percent until 2011 and 5.8 percent between 2012 and 2026 (Table 1). The projected increase reflects export diversification, significant improvements in physical infrastructure, and structural reforms to improve business environment and total factor productivity. Over the longer term, private consumption would be an important source of growth, due to higher incomes and continued inflows of remittances.

Inflation rate is projected to decline gradually to about 2½ percent by 2012 and stay at about that level thereafter. Monetary policy is geared toward achieving this objective.

Exports of goods and services are projected to grow (in US$ terms) at an annual rate of 13.4 percent during 2006-11, and at about 8 percent until 2026. In the long run, exports would be driven by nontraditional exports, which would grow at an annual rate of 16 percent between 2006 and 2026.

Remittances are assumed to be around 15 percent of annual GDP, while **foreign direct investment** is envisaged to average only 4 percent of GDP a year during 2007-11, and stay close to that share until 2026. **Official external grants** as a share of GDP are projected to decline slightly, while the **official concessional loans** are projected to average about 4 percent of GDP. The **reserve coverage of imports** would exceed 5 months of goods and services in the outer years.

The tax-to-GDP ratio would rise to 20.5 percent by 2012 as tax administration would strengthen—particularly in the areas where efficiency gains have not been exhausted, such as customs, exemptions, and the informal sector.

Structural reforms in PFM and SOEs would continue to improve governance and the efficiency of **public spending**. Increased mobilization of domestic revenue together with sustained external financing will allow the government to finance pro-poor and pro-growth spending, with a view to achieve the MDGs. The wage bill would be contained through civil service reform and limits on wage increase. The **overall budget deficit** is projected to stabilize at around 5.5 percent of GDP.

Table 1. Assumption on Key Variables

	Historical Average	Average 2006-11	Average 2012-26
Noninterest CAD, incl. transfers (percent of GDP)	3.9	7.4	5.5
Real GDP growth (percentage change)	4.7	7.0	5.8
Export growth (US dollar terms, percent)	10.0	13.4	7.9
Import growth (US dollar terms, percent)	13.1	13.7	8.9
Current transfers (official and private) to GDP (percent)	13.3	17.8	18.0

Borrowing Assumptions

Based on discussions with the authorities and development partners, it is assumed that gross concessional flows through 2012 do not exceed 7 percent of annual GDP, reflecting a stabilization after the peak observed in 2003–05. It is also assumed that nonconcessional borrowing would be in the range of US$200–350 million a year amounting to 1–1.7 percent of annual GDP between 2007 and 2012 (Table 2), and 1 percent of annual GDP afterwards.

Table 2. Gross Capital Flows, 2006–12
(Million of US$)

	2006		2007	2008	2009	2010	2011	2012
	Projected	Disbursed	Committed		Projected			
Concessional Flows	1,057	894	1,039	1,121	1,286	1,463	1,612	1,799
Percent of GDP	*8.2*	*6.9*	*6.9*	*6.6*	*6.7*	*6.7*	*6.6*	*6.7*
Nonconcessional Loans	0	0	250	250	200	350	350	300
Percent of GDP	*0*	*0*	*1.7*	*1.5*	*1.0*	*1.6*	*1.4*	*1.1*
Total Official Inflows	1,057	894	1,289	1,371	1,486	1,813	1,962	2,099
Percent of GDP	*8.2*	*6.9*	*8.6*	*8.1*	*7.7*	*8.3*	*8.1*	*7.8*

Assumptions on Interest Rates on International Markets

Assumptions about likely interest rates are based on Ghana's credit rating of B+ and conservatism on interest rate spreads. It is assumed that present favorable market conditions might not persist over the projection period, and therefore an annual interest rate of 9.5 percent is used.

Results

The results indicate that Ghana's risk of external debt distress is in the lower range of the moderate category. The 2005 World Bank Country Policy and Institutional Assessment (CPIA) classifies Ghana as a strong performer and assigns the following indicative debt-burden thresholds: (i) NPV of debt-to-exports of goods and services of 200 percent, (ii) NPV of debt-to-GDP of 50 percent, and (iii) debt service of 25 percent of exports of goods and services.[2] Under the baseline scenario, all of Ghana's external debt-burden indicators remain below these thresholds during the projection period, reflecting the fact that additional investment (including from nonconcessional sources) are expected to be growth enhancing.

[2] Bank and Fund staffs agreed to apply discretion and continue to use the latest annual CPIA observation for Ghana as a temporary exception to the three-year rule, to prevent "undue volatility in the performance rating" as in *Staff Guidance Note on the Application of the Joint Fund-Bank Sustainability Framework for Low-Income Countries* (www.imf.org).

The historical scenario is based on averages from 1996 to 2005, which includes a period of macroeconomic destabilization (large nominal depreciation, low export and import growth, etc. in 2000–01) and HIPC debt relief. In this scenario, all three debt burden indicators exceed their thresholds towards the end of the projection period. The results underscore the importance of continued macroeconomic stability while aiming for higher growth.

In the medium term, Ghana's debt burden trajectory breaches only one of the three commonly used indicative thresholds (the NPV of debt-to-GDP ratio) in only one of the six standardized stress tests, the *combined shock* (to real GDP growth, the GDP deflator, exports growth, and net current transfers). Under this test, Ghana's NPV of debt-to-GDP ratio rises from 18 percent in 2006 to 57 percent in 2008, staying above the 50 percent threshold until 2013, declining to about 40 percent thereafter.[3] The same shock would also significantly worsen the NPV of debt-to-exports ratio, which would reach 135 percent in 2008 and 142 percent in 2026. The deterioration of this indicator partly reflects a narrow export base and vulnerability to terms of trade shocks (Figure 1).

Table 3 illustrates the potential importance of remittance flows to debt sustainability. Remittance inflows are excluded from standard indicators of debt sustainability; their inclusion would significantly lower the indicators. For example, the NPV of debt-to-exports ratio would be about 40 percentage points of GDP lower in 2026 if remittances were included in foreign exchange earnings.[4]

Table 3. Debt Sustainability Indicators (baseline scenario)
(in percent)

	2006	2015	2026	Thresholds
NPV of debt-to-GDP	18	24	28	50
NPV of debt-to-exports	46	79	121	200
NPV of debt-to-exports and remittances	34	53	80	[200]
Debt service-to-exports	12	8	13	25
Debt service-to-exports and remittances	9	6	9	[25]

Source: IMF staff calculations.

[3] This shock adds a half-standard deviation worsening to historical values of the affected variables. Its impact is magnified in the case of Ghana by a sharp decline in the GDP deflator (leading to a higher import bill) plus much lower net current transfers, creating large current account deficits in 2007 and 2008. These deficits are then financed primarily at nonconcessional terms.

[4] Official remittance inflows are the fastest-growing source of foreign exchange in Ghana. As a share of GDP, remittances increased from 4 percent in 1995 to about 14 percent in 2006.

2. Alternative Scenarios

Alternative A—Baseline Borrowing, Lower GDP and Export Growth

This scenario considers what might happen if GDP and export growth do not accelerate. Specifically, it assumes lower growth of GDP (average annual growth of 5.3 percent until 2026) and exports (average annual growth of 8 percent until 2026).

On this macroeconomic path, debt distress for **Ghana would be moderate, but it would be notably higher than in the baseline.** All thresholds would be exceeded in the historical scenarios (Figure 2). The GDP-related threshold would be exceeded or reached throughout the entire the period if Ghana experienced a negative shock to GDP growth, exports, or to current flows.[5]

Table 4. Scenario A: Debt Sustainability Indicators (Baseline Borrowing/Low GDP and Export Growth)
(Percent)

	2006	2015	2026	Thresholds
NPV of debt-to-GDP	18	27	36	50
NPV of debt-to-exports	46	93	182	200
Debt service-to-exports	12	10	19	25

Alternative Scenario B—High Borrowing and Baseline GDP and Export Growth

This scenario assumes that would Ghana borrow US$500-600 million a year (2–3.5 percent of annual GDP) through 2012 on nonconcessional terms. It also assumes that these amounts would worsen the terms on which Ghana could borrow by 1 percentage point from all major creditors. All macroeconomic assumptions are same as in the baseline.

On this higher borrowing path and the baseline macroeconomic path, **Ghana would clearly be at moderate risk of debt distress**, but it would also be markedly higher than in the baseline scenario. All Ghana's external debt burden indicators are below their thresholds (Figure 3), but all historical scenarios breach the thresholds. The GDP-related threshold could be exceeded by a wide margin if Ghana suffered a combined negative shock to its GDP growth, exports, and FDI.[6]

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[5] These variables would be one-half standard deviation below the historical average in the next two years.

[6] These variables would be one-half standard deviation below the historical average in the next two years.

Table 5. Scenario B: Debt Sustainability Indicators (high borrowing/baseline growth) (in percent)				
	2006	2015	2026	Thresholds
NPV of debt-to-GDP	18	29	35	50
NPV of debt-to-exports	46	94	151	200
Debt service-to-exports	12	12	18	25

Alternative Scenario C—High Borrowing, Lower GDP and Export Growth

This scenario combines borrowing from alternative scenario B (high borrowing) and the macroframework from alternative scenario A (lower growth of GDP and exports). The results indicate that in this scenario, **Ghana would be approaching high risk of debt distress**, because two of the indicators are at or approaching their policy-dependent thresholds (NPV of debt -to-exports and debt service-to-exports), and all stress tests are above the thresholds. The results illustrate that exports (and foreign exchange) constitute one of the main risks for and constraints on debt sustainability, because the export base is relatively undiversified (Figure 4).

Table 6. Scenario C: Debt Sustainability Indicators (High Borrowing/Low growth scenario) (Percent)				
	2006	2015	2026	Thresholds
NPV of debt-to-GDP	18	30	39	50
NPV of debt-to-exports	46	101	199	200
Debt service-to-exports	12	14	24	25

B. Public Debt Sustainability Analysis

This section discusses the results of the public debt sustainability analysis (DSA) for Ghana using the new framework for low-income countries.

Main Results

The baseline scenario projects an increase in central government debt (external and domestic) from about 40 percent of GDP in 2007 to about 50 percent in 2026. The NPV of debt-to-GDP ratio would increase from about 38 percent in 2006 to about 47 percent in 2026, due to increased borrowing (Table B). Similarly, the NPV of debt-to-revenue ratio would rise from 140 percent in 2007 to about 181 percent in 2026. To check the robustness of the conclusions, we subject this scenario to a number of stress tests.

The sensitivity analyses suggest that the baseline scenario is most vulnerable to a shock to real GDP growth. If real GDP growth were to stay at historical average minus one standard deviation in 2008–09, the NPV of debt-to-GDP ratio would increase by

18 percentage points in 2026, while the NPV of debt-to-revenue ratio would rise by 69 percentage points in 2026 (relative to the baseline).

A shock that maintains the primary deficit at its peak 2006 level would also have a significant impact on the debt dynamics, raising the NPV of debt-to-GDP ratio by about 14 percentage points. This is not surprising as the 2006 primary deficit has been the worst since 1995.

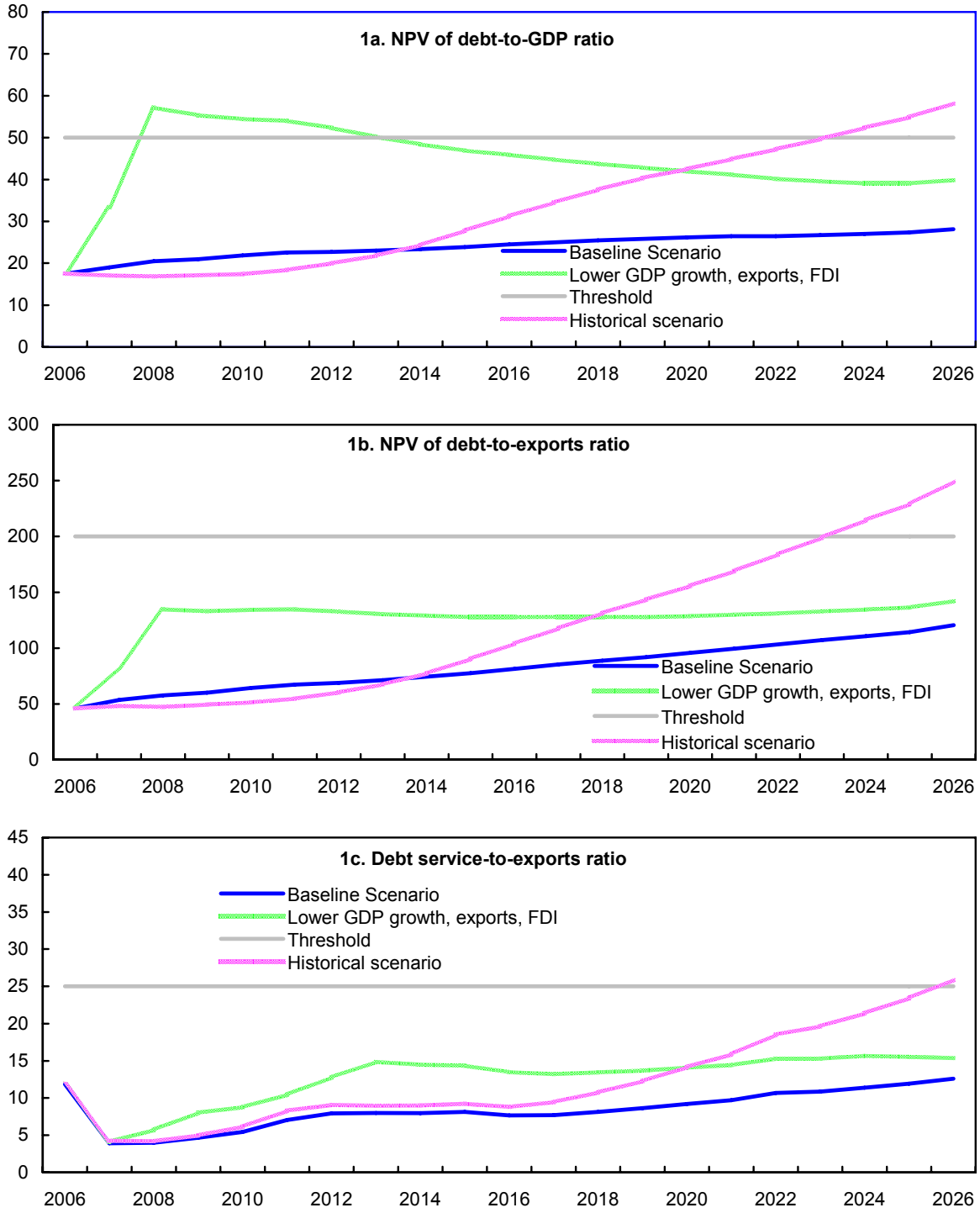
The baseline scenario remains vulnerable to contingent liabilities. While reliable and comprehensive data on contingent liabilities are not available, a stress could help gauge the likely impact of these flows on the debt profile. For example, a 10-percent increase in debt/ GDP resulting from contingent liabilities in 2008 would increase the NPV of debt-to-GDP ratio by 9 percentage points in 2008 and by about 3 percentage points in 2026. It also would increase the NPV of debt-to-revenue ratio by 36 percentage points in 2008 and by 12 percentage points in 2026.

Overall, the public debt analysis points to moderate risk of debt distress. Combining (nominal) external and domestic debt shows that Ghana's total public debt would increase from about 40 percent of GDP in 2007 to about 50 percent in 2026 under the baseline scenario.

Summary

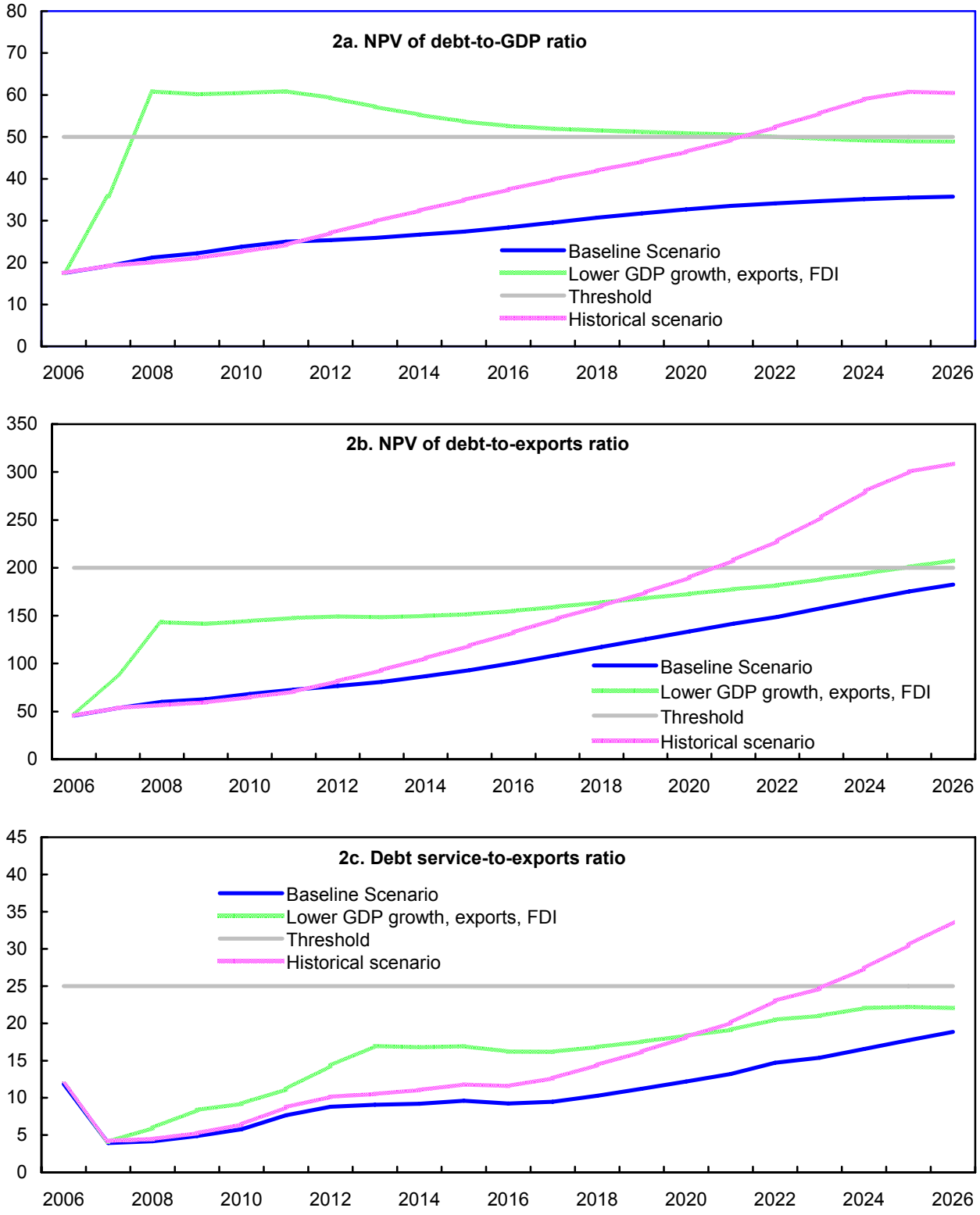
The conclusion of the DSA is that Ghana's risk of external debt distress is moderate, albeit close to the low risk category. Even under country-specific alternative scenarios, the baseline debt trajectory stays below the relevant indicative thresholds**.** The results of the country-specific alternative scenarios also indicate, however, that sustainability could deteriorate in the case of slower than forecast growth and/or increased borrowing on nonconcessional terms. **The risks to total public debt are also moderate**. These DSA results underscore the need for the authorities to maintain sound macroeconomic policies and accelerate structural reforms to achieve higher GDP growth and diversify exports. While these measures would reduce Ghana's vulnerability to exogenous shocks, staying on a sustainable path would also require that Ghana continues with prudent debt management, accesses international markets in a gradual and cautious manner, and uses all borrowed funds judiciously.

Figure 1. Ghana: Indicators of Public and Publicly Guaranteed External Debt -- Baseline Scenario, 2006–2026
(Percent)



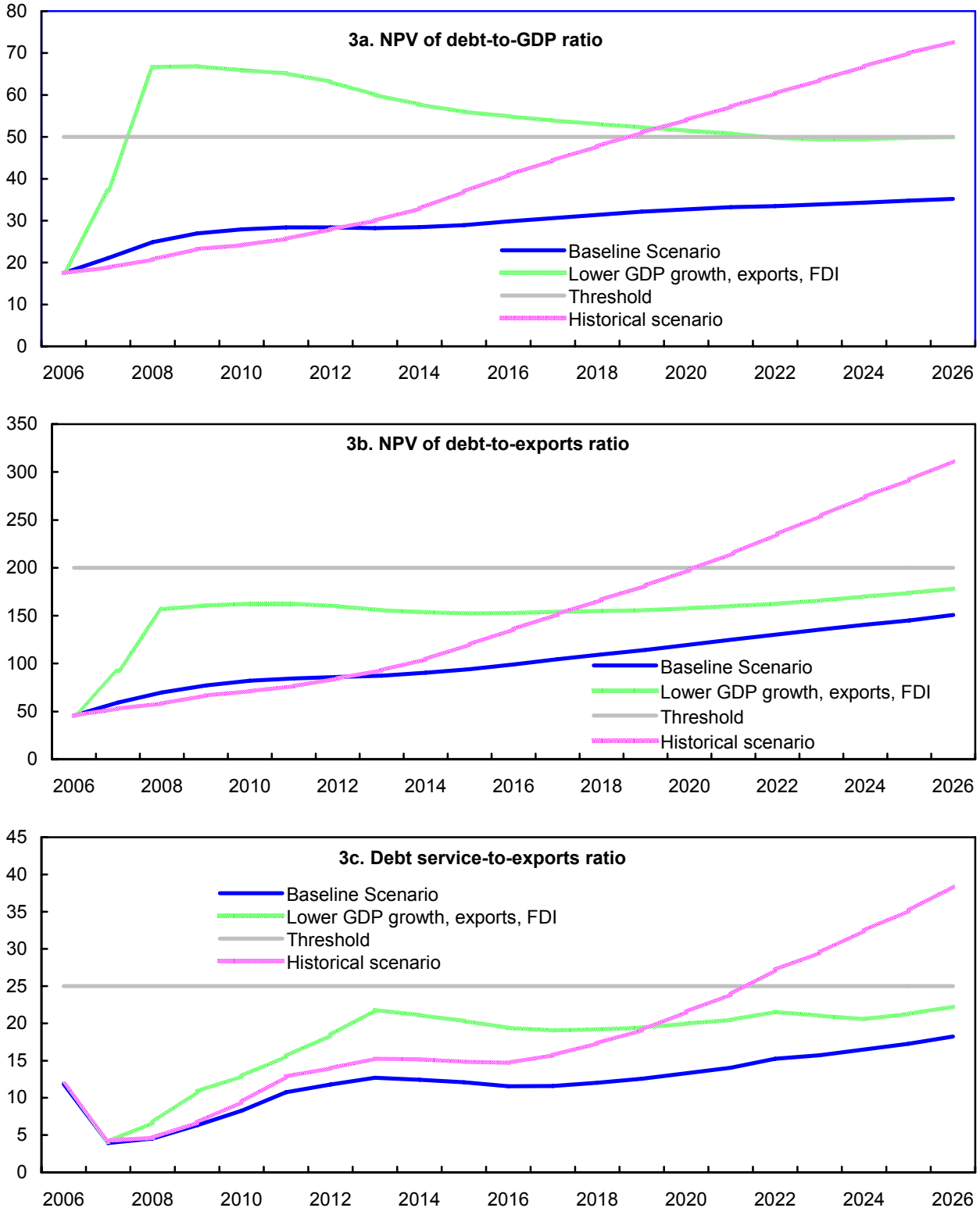
Sources: IMF staff projections and simulations.

Figure 2. Ghana: Indicators of External Debt -- Alternative Scenario A (Baseline Borrowing with Low GDP and Export Growth), 2006–2026
(Percent)



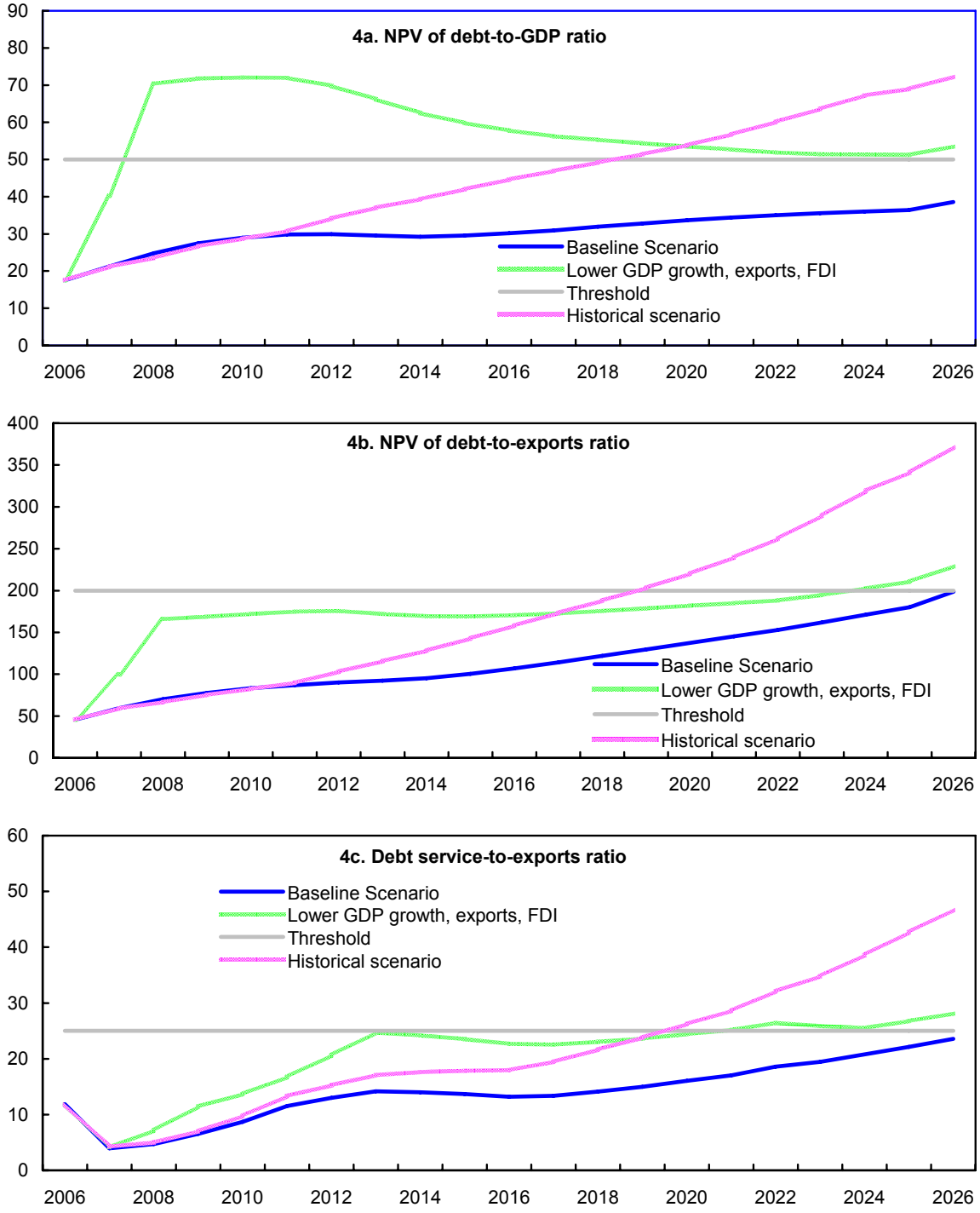
Sources: IMF staff projections and simulations.

Figure 3. Ghana: Indicators of External Debt -- Alternative Scenario B (High Borrowing with Baseline GDP and Export Growth), 2006–2026
(Percent)



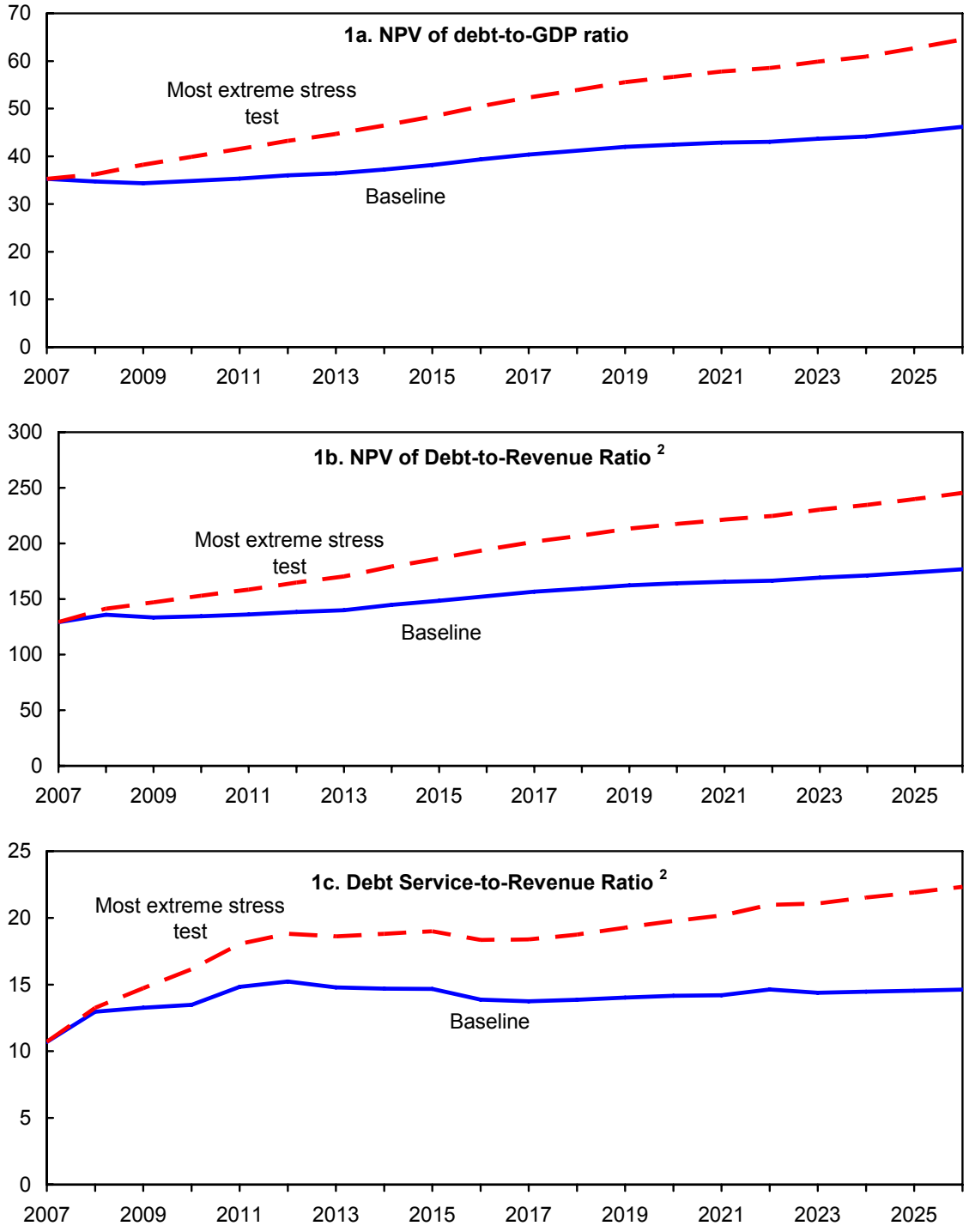
Sources: IMF staff projections and simulations.

Figure 4. Ghana: Indicators of External Debt -- Alternative Scenario C (High Borrowing with Low GDP and Export Growth), 2006–2026
(Percent)



Sources: IMF staff projections and simulations.

Figure 5. Ghana: Indicators of Public Debt Under Alternative Scenarios, 2007-2026[1]



Source: IMF staff projections and simulations.

[1] Most extreme stress test is test that yields highest ratio in 2026.

[2] Revenue including grants.

Table A. Ghana: External Debt Sustainability Framework, Baseline Scenario, 2003-2026 [1]
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average [6]	Standard Deviation [6]	Projections									
	2003	2004	2005			2006	2007	2008	2009	2010	2011	2006-11 Average	2012	2026	2012-26 Average
External debt (nominal) [1]	**99.0**	**72.7**	**59.2**			**22.3**	**23.6**	**25.3**	**25.8**	**26.6**	**27.3**		**27.5**	**31.1**	29.6
o/w public and publicly guaranteed (PPG)	99.0	72.7	59.2			22.3	23.6	25.3	25.8	26.6	27.3		27.5	31.1	29.6
Change in external debt	-0.5	-26.3	-13.5			-36.9	1.3	1.7	0.5	0.8	0.7		0.2	0.3	
Identified net debt-creating flows	-21.8	-15.0	-6.9			3.2	5.0	4.3	2.8	3.3	1.8		0.7	0.0	
Non-interest current account deficit	**-3.2**	**0.7**	**5.4**	**3.9**	**5.2**	**7.6**	**8.1**	**7.5**	**6.4**	**7.7**	**7.0**	**7.4**	**5.9**	**5.1**	5.5
Deficit in balance of goods and services	12.0	21.1	25.6			24.1	25.6	25.3	24.5	25.4	25.0		24.0	23.7	
Exports	40.7	39.3	36.1			38.2	35.5	35.6	34.9	34.0	33.6		33.1	23.4	
Imports	52.7	60.4	61.7			62.3	61.0	60.9	59.3	59.4	58.6		57.1	47.0	
Net current transfers (negative = inflow)	-15.7	-20.6	-19.7	-13.3	4.9	-16.9	-17.6	-17.8	-18.2	-18.2	-18.2	-17.8	-18.2	-17.7	-18.0
Other current account flows (negative = net inflow)	0.4	0.3	-0.4			0.4	0.1	0.0	0.1	0.5	0.2		0.1	-0.9	
Net FDI (negative = inflow)	**-1.0**	**-3.7**	**-1.4**	**-1.4**	**1.0**	**-1.9**	**-2.3**	**-2.4**	**-2.7**	**-3.4**	**-4.2**	**-2.8**	**-4.3**	**-4.7**	-4.6
Endogenous debt dynamics [2]	**-17.6**	**-12.0**	**-10.9**			**-2.5**	**-0.7**	**-0.8**	**-0.9**	**-1.0**	**-1.0**		**-0.9**	**-0.5**	
Contribution from nominal interest rate	1.5	1.9	1.6			0.6	0.5	0.6	0.7	0.7	0.8		0.9	1.0	
Contribution from real GDP growth	-4.2	-4.8	-3.5			-3.1	-1.2	-1.4	-1.7	-1.7	-1.8		-1.8	-1.5	
Contribution from price and exchange rate changes	-14.9	-9.2	-9.0			…	…	…	…	…	…		…	…	
Residual (3-4) [3]	**21.3**	**-11.3**	**-6.6**			**-40.0**	**-3.8**	**-2.6**	**-2.3**	**-2.5**	**-1.0**		**-0.5**	**0.3**	
o/w exceptional financing	-3.8	-4.4	-3.3			-0.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt [4]	...	...	19.9			17.5	19.0	20.5	20.9	21.9	22.6		22.7	28.1	
In percent of exports	...	...	55.1			45.9	53.6	57.7	60.1	64.4	67.2		68.7	120.5	
NPV of PPG external debt	**...**	**...**	**19.9**			**17.5**	**19.0**	**20.5**	**20.9**	**21.9**	**22.6**		**22.7**	**28.1**	
In percent of exports	...	...	55.1			45.9	53.6	57.7	60.1	64.4	67.2		68.7	120.5	
Debt service-to-exports ratio (in percent)	14.5	16.2	15.3			11.9	3.9	4.0	4.7	5.5	7.1		7.9	12.6	
PPG debt service-to-exports ratio (in percent)	14.5	16.2	15.3			11.9	3.9	4.0	4.7	5.5	7.1		7.9	12.6	
Total gross financing need (billions of U.S. dollars)	0.1	0.3	1.0			1.3	1.2	1.3	1.2	1.5	1.5		1.4	4.7	
Non-interest current account deficit that stabilizes debt ratio	-2.7	27.0	18.8			44.5	6.8	5.8	5.9	6.9	6.2		5.7	4.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.2	5.6	5.9	4.7	0.7	6.2	6.3	6.9	7.5	7.5	7.5	7.0	7.3	5.3	5.8
GDP deflator in US dollar terms (change in percent)	17.6	10.2	14.1	1.8	15.6	13.2	9.3	5.3	6.2	5.9	3.1	7.2	3.2	3.2	4.5
Effective interest rate (percent) [5]	1.9	2.3	2.7	2.3	0.4	1.2	2.6	3.0	3.2	3.3	3.4	2.8	3.5	3.7	3.5
Growth of exports of G&S (US dollar terms, in percent)	18.7	12.4	11.0	10.0	13.1	27.3	7.9	12.9	11.9	10.8	9.7	13.4	9.0	6.5	7.9
Growth of imports of G&S (US dollar terms, in percent)	19.1	33.2	23.4	13.1	16.6	21.6	13.8	12.2	11.4	13.9	9.3	13.7	7.9	9.4	8.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	31.7	21.3	20.9	22.4	17.3	17.2	21.8	18.3	12.8	14.8
Memorandym item:															
Nominal GDP (billions of US dollars)	7.6	8.9	10.7			12.9	15.0	16.9	19.3	21.9	24.3		26.9	109.1	

Source: Staff simulations.

[1] Includes both public and private sector external debt.

[2] Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

[3] Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

[4] Assumes that NPV of private sector debt is equivalent to its face value.

[5] Current-year interest payments devided by previous period debt stock.

[6] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table B. Ghana: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-2026
(In percent of GDP, unless otherwise indicated)

	Actual					Projections													
	2004	2005	2006	Historical Average [5]	Standard Deviation [5]	2007	2008	2009	2010	2011	2012	2015	2020	2021	2022	2023	2024	2025	2026
Public sector debt [1]	93.4	77.1	42.4			39.6	39.1	39.2	39.2	39.8	40.6	42.9	46.6	46.8	47.0	47.3	47.8	48.6	50.2
o/w foreign-currency denominated	72.7	59.2	22.3			23.6	25.3	25.8	26.6	27.3	27.5	28.6	30.2	30.4	30.3	30.4	30.6	30.8	31.1
Change in public sector debt	-25.4	-16.3	-34.7			-2.7	-0.6	0.2	-0.1	0.6	0.8	0.8	0.4	0.2	0.2	0.4	0.5	0.8	1.7
Identified debt-creating flows	-14.1	-13.5	-5.6			-1.5	0.3	1.0	0.6	1.3	1.4	0.9	0.4	0.2	0.2	0.4	0.5	0.8	0.8
Primary deficit	-0.8	-0.7	4.2	1.4	2.1	4.4	3.5	3.5	3.1	3.1	3.2	2.7	2.8	2.7	2.7	2.8	2.8	2.9	3.0
Revenue and grants	30.1	29.1	26.8			27.3	25.6	25.8	25.9	26.0	26.0	25.7	25.9	25.9	25.9	25.8	25.8	26.0	26.1
of which: grants	6.4	5.2	5.2			5.1	3.4	3.5	3.5	3.4	3.4	3.1	2.9	2.9	2.9	2.8	2.8	2.7	2.7
Primary (noninterest) expenditure	29.3	28.4	31.1			31.7	29.1	29.3	29.0	29.1	29.3	28.4	28.7	28.7	28.6	28.6	28.6	28.9	29.1
Automatic debt dynamics	-12.9	-12.6	-9.9			-3.8	-2.3	-2.5	-2.5	-1.8	-1.8	-1.8	-2.4	-2.5	-2.5	-2.4	-2.4	-2.1	-2.2
Contribution from interest rate/growth differential	-6.1	-5.6	-3.9			-2.3	-1.4	-1.5	-1.7	-1.6	-1.5	-1.3	-1.5	-1.6	-1.6	-1.6	-1.7	-1.6	-1.7
of which: contribution from average real interest rate	0.2	-0.4	0.6			0.2	1.1	1.2	1.1	1.1	1.2	1.2	0.8	0.7	0.8	0.7	0.7	0.8	0.7
of which: contribution from real GDP growth	-6.3	-5.2	-4.5			-2.5	-2.5	-2.7	-2.7	-2.7	-2.7	-2.4	-2.3	-2.4	-2.4	-2.4	-2.4	-2.4	-2.4
Contribution from real exchange rate depreciation	-6.9	-7.0	-6.0			-1.5	-0.9	-1.0	-0.9	-0.3	-0.3	-0.6	-0.9	-0.9	-1.0	-0.8	-0.7	-0.5	-0.5
Other identified debt-creating flows	-0.4	-0.2	0.0			-2.1	-0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Privatization receipts (negative)	-0.4	-0.2	0.0			-2.1	-0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	-11.3	-2.8	-29.1			-1.2	-0.9	-0.8	-0.7	-0.6	-0.6	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.9
NPV of public sector debt	...	...	37.6			35.0	34.3	34.4	34.4	35.0	35.8	38.1	42.5	42.9	43.1	43.6	44.2	45.1	47.2
o/w foreign-currency denominated	...	...	17.5			19.0	20.5	20.9	21.9	22.6	22.7	23.8	26.1	26.4	26.5	26.7	27.0	27.4	28.1
o/w external	…	…	17.5			19.0	20.5	20.9	21.9	22.6	22.7	23.8	26.1	26.4	26.5	26.7	27.0	27.4	28.1
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...	...	...	...	...	...	...	...	...
Gross financing need [2/]	6.0	5.1	9.4			7.4	6.9	6.9	6.6	7.0	7.2	6.5	6.5	6.4	6.5	6.5	6.6	6.7	6.8
NPV of public sector debt-to-revenue ratio (in percent) [3]	…	…	140.0			128.4	134.1	133.6	133.0	134.8	137.6	148.2	164.3	165.3	166.5	169.0	171.5	173.8	180.8
o/w external	…	…	65.3			69.6	80.2	81.3	84.5	86.8	87.3	92.6	101.0	101.9	102.3	103.5	104.6	105.4	107.7
Debt service-to-revenue ratio (in percent) [3 4]	22.4	20.0	19.3			10.7	12.9	13.3	13.5	14.8	15.2	14.7	14.2	14.2	14.6	14.4	14.5	14.5	14.6
Primary deficit that stabilizes the debt-to-GDP ratio	24.6	15.6	39.0			7.2	4.1	3.3	3.2	2.5	2.4	1.9	2.4	2.5	2.5	2.4	2.4	2.1	1.3
Key macroeconomic and fiscal assumptions																			
Real GDP growth (in percent)	5.6	5.9	6.2	4.9	0.8	6.3	6.9	7.5	7.5	7.5	7.3	6.1	5.3	5.3	5.3	5.3	5.3	5.3	5.3
Inflation rate (GDP deflator, in percent)	14.4	15.0	14.6	20.8	7.2	11.2	9.0	8.2	7.4	6.5	5.5	4.2	5.3	5.4	5.4	4.7	4.5	3.8	3.8
Growth of real primary spending (deflated by GDP deflator, in percent)	30.9	2.3	16.4	8.1	13.7	8.6	-1.9	8.1	6.6	7.8	7.8	5.9	5.1	5.1	5.1	5.3	5.5	6.2	6.1

Sources: Country authorities; and Fund staff estimates and projections.

[1] Coverage of public sector is central government; gross debt is used.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues including grants.

[4] Debt service is defined as the sum of interest and amortization of medium and long-term debt.

[5] Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.